Disclosure Before Trust: Subgroup-Aware Ethical Transparency and the Calibration of Reliance in Human–AI Decision Making

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***Abstract*—Calls for ethical AI are now abundant, but it is far less clear how an ethical commitment such as transparency actually reaches the person who has to decide, case by case, whether to follow an AI recommendation. This paper treats trust as a quantity to be calibrated rather than maximised, and asks whether the form of ethical disclosure changes how well human reliance tracks real AI performance for different groups of people. We propose a three-layer framework that links an ethics layer (subgroup audit), a disclosure layer and a reliance layer, and we define the Ethical Trust Calibration Index (ETCI), which combines appropriate reliance with parity between groups. In an agent-based simulation of 300 runs of 2,000 decisions across five conditions, aggregate accuracy disclosure and outcome feedback left a minority group at 0.62–0.65 joint accuracy when the AI was right 62% of the time for that group, whereas a subgroup-aware disclosure card raised it to 0.78, equal to unaided humans, and lifted ETCI from 0.49 to 0.86. The card did not remove the gap between groups (0.14), which shows that transparency limits harm but cannot stand in for model remediation. The results come from a simulation with stated assumptions, and we close with a protocol for testing them on human participants.**

***Index Terms*—AI ethics, human–AI trust, trust calibration, algorithmic transparency, subgroup fairness, responsible AI.**

I. INTRODUCTION

An AI system that is accurate on average can still be unreliable for the people it serves least well. Face analysis tools, clinical risk scores and screening models have all been shown to perform unevenly across demographic groups [5], [6]. When such a system is placed next to a human decision maker, the harm depends on something that is rarely measured: whether the human relies on the AI where it is good and relies on themselves where it is not. Ethics guidelines tell developers to be transparent and tell users to be vigilant [1], [2], but they say little about the point where these two meet, which is the moment a person looks at a recommendation and decides what to do with it.

This paper focuses on that moment. Our starting position is that the ethical goal in human–AI teams is calibrated trust, not high trust. A person who follows an AI everywhere is as badly served as one who follows it nowhere. What the person needs is information that lets reliance vary with the actual reliability of the AI in the case at hand [3], [4]. If that information is an overall accuracy figure, it will be wrong for every group whose accuracy differs from the average, and the people in those groups bear the cost.

We examine three questions. RQ1: does the granularity of ethical disclosure (none, aggregate, or per subgroup) change how well reliance matches AI performance? RQ2: what does this do to the accuracy and the gap between groups? RQ3: can outcome feedback substitute for a proper audit? Our contributions are as follows.

(1) A three-layer framework connecting subgroup auditing, disclosure and reliance, with a short argument for why aggregate disclosure cannot be sufficient (Proposition 1). **(2)** The Ethical Trust Calibration Index, a single measure that penalises both miscalibrated reliance and unequal outcomes. **(3)** A reproducible simulation comparing five disclosure conditions, together with a sensitivity analysis and a validation protocol for human-subject work.

II. BACKGROUND AND RELATED WORK

A. Ethical principles and their operational gap

Reviews of published guidelines find convergence on a small set of principles, with transparency, justice and fairness, non-maleficence, responsibility and privacy most frequently named [1]. Floridi et al. distilled a comparable set into five principles, including explicability [2]. Regulation is following: the EU Artificial Intelligence Act places information duties on providers of high-risk systems so that deployers can interpret and appropriately use outputs [18]. What these texts leave open is the form the information should take. Fairness criteria such as individual fairness and equality of opportunity [11], [12] define what equal treatment means for a model, yet they say nothing about what the human who uses the model should be told.

B. Disclosure artefacts

Model cards, datasheets and internal audit practice were proposed to make performance and limits visible [7], [14], [15]. Model cards in particular ask for results disaggregated by group. Their intended readers are mostly developers, regulators and procurement teams. How such disaggregated information changes the behaviour of a frontline user at the point of decision has received much less attention, and that is the gap we address.

C. Trust, reliance and calibration

The automation literature distinguishes trust as an attitude from reliance as a behaviour, and defines appropriate reliance as reliance that matches the true capability of the system [3], [4], [16]. Trust is usually analysed through ability, integrity and benevolence of the trustee [17]; for an AI tool in a decision loop, ability, meaning its measurable accuracy, is the dimension that disclosure can most directly inform. Empirical work shows that people can abandon an algorithm after seeing it err [10], that displayed confidence can shift reliance when it is itself well calibrated [13], and that explanations can increase acceptance of AI advice even when the advice is wrong [9]. Modern neural classifiers are often overconfident in the first place [8], so a displayed confidence score is a weak foundation for trust. None of this work asks whether the same disclosure is appropriate for every subgroup of the people being decided about.

III. FRAMEWORK

Fig. 1 summarises the framework. The ethics layer produces evidence: an audit that estimates the accuracy α_g of the AI for each relevant group g, together with stated limits of use. The disclosure layer decides which part of that evidence reaches the user at decision time. The reliance layer is the person. From what they have been shown they form a belief b about how accurate the AI is for the case in front of them, compare it with their belief h about their own accuracy, and either follow the AI or decide alone. Outcomes then return, late and only sometimes, as feedback that updates the belief.

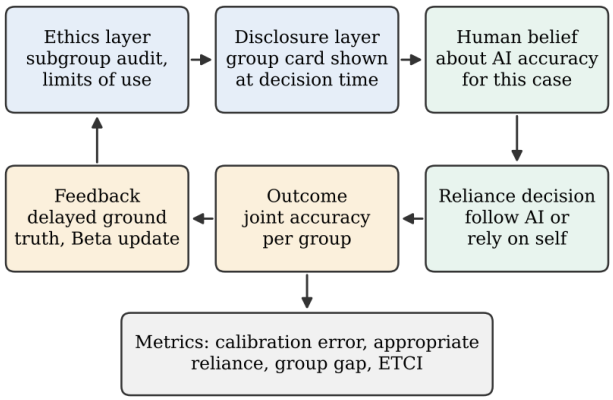


Fig. 1. Three-layer framework. Evidence from an ethics audit is filtered by a disclosure layer into a belief, which drives reliance; outcomes feed back to the audit and to the belief, and four metrics are computed on the result.

A. Appropriate reliance

Let a_t denote the true accuracy of the AI for the group of case t. Reliance on case t is appropriate if the person follows the AI exactly when $a_t > h$. This treats AI and human errors as independent, which is a simplification we return to in Section VII. The user rule used throughout is: follow the AI if $b + \varepsilon > h$, where ε is zero-mean noise representing inconsistent judgement.

Proposition 1. Suppose disclosure conveys a single number $b = \bar{\alpha}$ (overall accuracy) for every case. Then the decision rule treats all groups alike, so reliance is appropriate for every group only if $\alpha_g - h$ has the same sign for all g. In particular, if $\bar{\alpha} > h$ and some group has $\alpha_g < h$, that group is over-relied upon, and if $\bar{\alpha} < h$ and some group has $\alpha_g > h$, that group is under-served. The argument is immediate from the rule, since b does not depend on g. The practical point is that average accuracy is exactly the statistic that is most likely to look reassuring while hiding a subgroup failure.

B. Metrics

We use three measures. Calibration error is the mean distance between belief and truth, and appropriate reliance is the share of decisions that follow the rule above:

$$CE = (1/T)\,\Sigma_t \,|\, b_t - a_t \,| \qquad (1)$$

$$AR = (1/T)\,\Sigma_t \, 1[\ \text{follow}_t = (a_t > h)\ ] \qquad (2)$$

The Ethical Trust Calibration Index combines appropriate reliance with parity between the two groups, whose joint accuracies are Acc_A and Acc_B:

$$\mathrm{ETCI} = \mathrm{AR} \times (\,1 - |\,\mathrm{Acc}A - \mathrm{Acc}B\,|\,) \qquad (3)$$

ETCI lies in [0, 1] and equals 1 only when reliance is always appropriate and both groups achieve the same accuracy. The multiplicative form is a design choice: a system cannot compensate for unequal outcomes with good calibration, or the reverse. ETCI can be raised by lowering the better-off group, so we always report group accuracies next to it.

IV. SIMULATION DESIGN

We use an agent-based simulation because it lets us fix the true accuracy of each group, which is unobservable in field data, and then vary only the disclosure. Each run presents T = 2,000 binary decisions. A case belongs to the minority group B with probability 0.30. The AI is correct with probability 0.92 for group A and 0.62 for group B; the unaided human is correct with probability 0.78 for both groups; the two are drawn independently. Overall AI accuracy is therefore 0.83, above the human baseline, which is the situation in which an average-only report looks reassuring. Table I lists all parameters.

TABLE I
SIMULATION PARAMETERS

Parameter	Value
Decisions per run T / runs	2,000 / 300
Share of group B	0.30
AI accuracy, group A / group B	0.92 / 0.62
Unaided human accuracy h	0.78
Displayed AI confidence (no disclosure)	0.90
Judgement noise, s.d. of ε	0.05
Audit sample per group (card)	200 cases
Feedback probability / prior strength	0.30 / 20

Five conditions differ in what the user believes about the AI. (C1) No disclosure: the user sees only the confidence score, fixed at 0.90 to reflect the overconfidence typical of neural models [8]. (C2) Aggregate disclosure: the user is told overall accuracy. (C3) Group card: the user is told the accuracy of the AI for the group of the current case, estimated from an audit of 200 cases per group, so that it carries sampling error. (C4) No disclosure with feedback: the user starts from 0.90 as a Beta prior of strength 20 and updates a single global belief whenever feedback arrives, which happens on 30% of cases. (C5) Group card with feedback: the same updating, but with one belief per group starting from the audited value.

For each condition we ran 300 independent replications and report means with 95% normal-approximation intervals. For the sensitivity analysis we varied the accuracy of the AI on group B from 0.55 to 0.90 with 100 replications per point. The code is about a hundred lines of Python and NumPy with a fixed seed (2027) and is available from the author on request.

V. RESULTS

TABLE II
MAIN RESULTS (MEAN OF 300 RUNS)

Condition	Acc	Acc A	Acc B	Gap	AR	ETCI
Human alone	0.780	0.780	0.780	0.000	–	–
AI alone	0.830	0.920	0.620	0.300	–	–
C1 None	0.830	0.919	0.622	0.297	0.696	0.489
C2 Aggregate	0.822	0.897	0.647	0.250	0.636	0.477
C3 Card	0.878	0.920	0.779	0.141	0.995	0.855
C4 None+FB	0.822	0.898	0.643	0.255	0.636	0.473
C5 Card+FB	0.877	0.919	0.777	0.143	0.994	0.852

Half-widths of the 95% intervals are below 0.01 for every figure in Table II, so differences discussed below are not sampling artefacts of the simulation, although they remain conditional on its assumptions.

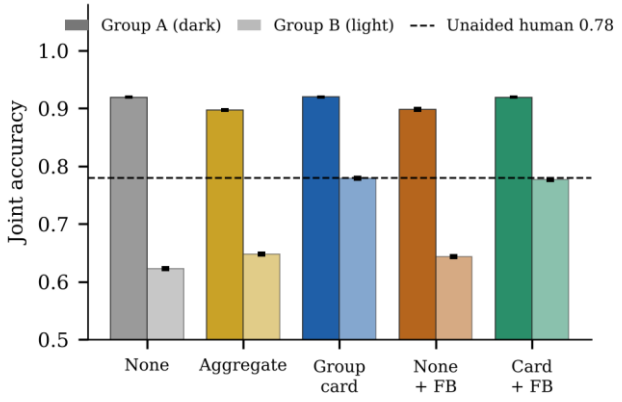


Fig. 2. Joint accuracy by group and condition. Only the group card brings group B up to the unaided human level (dashed line). Error bars are 95% intervals.

Without disclosure, the average hides the failure. Users followed the AI on 99.2% of group B cases. Group B accuracy was 0.622, which is 0.16 below what the same people achieve alone and essentially equal to the AI on its own (0.620). The overall figure of 0.830 looks acceptable and conceals this entirely.

Aggregate disclosure moves the error rather than removing it. Telling users that the AI is 83% accurate put their belief close to the human baseline, so noise decided each case: reliance fell to 84% for both groups. Group B gained 0.025 in accuracy, but group A lost 0.022, overall accuracy fell from 0.830 to 0.822, and AR fell from 0.696 to 0.636, because users now under-used the AI where it was good.

The group card repairs reliance. With subgroup information, reliance on group B dropped to 0.4% while remaining at 99.5% for group A. Group B accuracy rose to 0.779, matching the unaided baseline, group A kept 0.920, overall accuracy reached 0.878, calibration error fell from 0.098 to 0.019, AR reached 0.995, and ETCI rose from 0.489 to 0.855.

Feedback does not replace an audit. A user who learns only a global belief converges to overall accuracy, which is what aggregate disclosure supplies, and the results of C4 are almost identical to C2 (ETCI 0.473 vs. 0.477). Calibration error on group B stayed at about 0.21 after 2,000 decisions (Fig. 4). Adding feedback to the card made no measurable difference to the final outcome (ETCI 0.852 vs. 0.855).

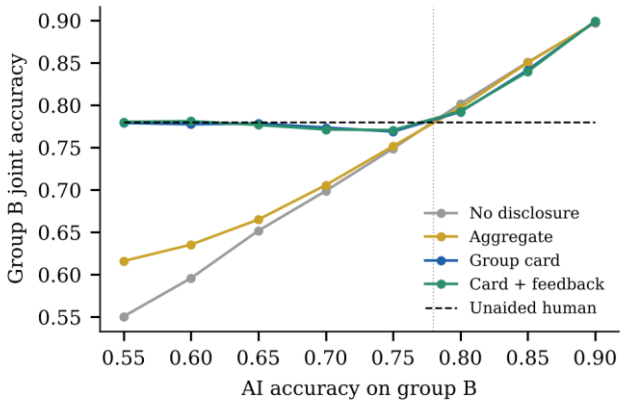


Fig. 3. Group B accuracy as the AI's accuracy on that group varies. Below the human baseline (0.78) only the card protects group B; above it the card costs about 0.01 because of audit noise.

A. Sensitivity to the size of the disparity

Fig. 3 shows that the benefit depends on how far group B lies below the human baseline. With the AI at 0.55 on group B, users without disclosure reach 0.551 against 0.779 with the card. The card curve is flat at about 0.77–0.78 until the AI overtakes humans on that group, after which the curves cross. Above the crossing the card costs up to about 0.01 (0.792 against 0.802 at 0.80, and 0.842 against 0.851 at 0.85), because a noisy audit sometimes tells a user to rely on themselves when the AI would have been better.

This cost is predictable. If the audit estimate has standard deviation $\sqrt{(\alpha(1-\alpha)/n)}$ and the user noise has deviation σ, a group whose true accuracy lies δ from the human baseline is mishandled with probability

$$P\text{err} = \Phi\left(-\delta / \sqrt{\left(\alpha(1-\alpha)/n + \sigma^2 \right)} \right) \qquad (4)$$

where Φ is the standard normal distribution function. For group B in the main experiment ($\delta = 0.16$, $n = 200$, $\sigma = 0.05$) this gives about 0.4%, in line with the observed reliance of 0.4%. For a group within 0.02 of the baseline it gives about 36%, which explains the dip in appropriate reliance near the crossing in Fig. 3.

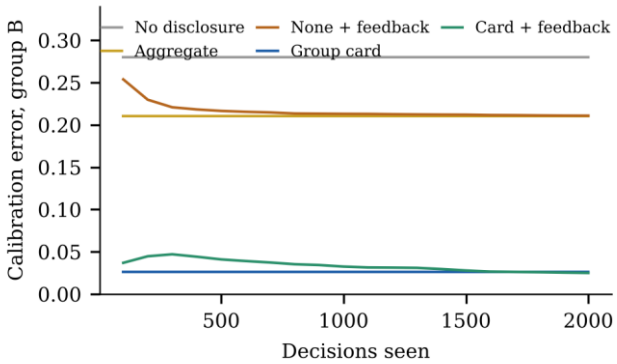


Fig. 4. Calibration error on group B across the first 2,000 decisions. Users without group information settle at the error of the global average; the card is accurate from the start.

VI. DISCUSSION

A. Granularity matters more than volume

The three disclosure conditions give the user the same amount of information in a loose sense: one number. What changed the outcome was which number. A global figure and a displayed confidence score are both indifferent to who the case is about, and Proposition 1 shows why this cannot be fixed by making the figure more accurate. For ethics guidance this means that "be transparent" is under-specified. A defensible reading is that transparency should be disaggregated along the dimensions on which the system is known or suspected to perform unevenly.

B. Transparency limits harm but does not remove inequity

Even under the card, the gap between the groups remained at 0.141. Group B users are returned to unaided judgement at 0.78 while group A users enjoy 0.92 with the AI. The card prevents the AI from making group B worse off than no AI at all, but it does not make group B better off. This is a useful boundary for the ethics of disclosure: it should not be offered as a reason to deploy a model that is poor for some group. Closing the gap requires work on the model, such as more representative data, or on the process, for example more human support for the affected group.

C. Audits are themselves unevenly accurate

Eq. (4) points to a less obvious fairness issue inside the mechanism. Audit error grows as group size shrinks, so the groups with the least data receive the noisiest cards and are the most likely to be mishandled when they are close to the human baseline. We therefore suggest that a card report the audit size and an interval with every group figure, and that minimum audit sizes be set by the size of the disparity one needs to detect.

D. Practical implications

Our results suggest five practices for a team deploying an AI recommender: show subgroup accuracy at the decision screen rather than in a separate report; give audit sizes and intervals; refresh the audit when the population shifts; monitor group accuracies and ETCI after deployment; and settle in advance what happens for a group where the AI falls below the human baseline, whether that is deferral, withholding the AI output, or retraining.

VII. LIMITATIONS AND VALIDATION PLAN

The findings are properties of a model, not measurements of people. The simulated user follows a threshold rule, knows their own accuracy, and is fully informed about group membership. Real users show automation bias, anchoring, and algorithm aversion after visible errors [10], and human and AI errors are often correlated, which would change what appropriate reliance means. Accuracies are static, there are only two groups with no intersectional structure, costs of errors are symmetric, and the main result uses one parameter setting, with sensitivity examined only for the accuracy on group B. The weighting inside ETCI is a design choice. The simulation therefore demonstrates a mechanism and its direction, not an effect size.

To test the mechanism we propose a pre-registered between-subjects experiment. Participants would make classification decisions with an AI assistant whose accuracy is controlled by group, under the arms C1 to C3 plus a feedback arm. The primary outcome is appropriate reliance by group (H1: AR for minority cases is higher under the card than under aggregate disclosure, and higher under aggregate than under none). Secondary outcomes are calibration error, a validated self-report trust scale, decision time and perceived fairness. The sample should be sized by power analysis for a medium effect, and the study needs ethics review.

VIII. CONCLUSION

Ethical principles reach users through disclosure, and the granularity of disclosure determines whether trust is calibrated. In our simulation, aggregate accuracy and outcome feedback both left a minority group relying on an AI that was wrong for it 38% of the time, while a subgroup-aware card removed that harm and raised the Ethical Trust Calibration Index from 0.49 to 0.86. The card did not remove the disparity itself, so disclosure belongs beside, not in place of, model remediation. The next step is empirical: testing these predictions with human participants, and extending the framework to correlated errors, intersectional groups and explanations.

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